

COLOUR BUDDY: Learn Colours With Lights

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This system uses a colour-changing LED (RGB LED/ tri-colour LED) that switches to a new colour every 5 seconds. Simultaneously, the colour name is displayed on an OLED mini screen (SSD1306). This pairing helps learners connect colour names with the actual shades, making the process engaging and enjoyable.

Colour Buddy is a playful yet educational system that merges technology with early childhood learning. It highlights how simple electronics can be turned into interactive learning aids for children. Fig. 1 shows the author's prototype. The Bill of Materials required to build the system is listed in Table 1.

This circuit is designed for children. It makes learning colours enjoyable by combining visual display and text, while also introducing beginners to microcontrollers, LED colour mixing, and I²C display control.

The idea is based on the principle that two different colour combinations can generate a new colour. For example, by adding red and green, yellow is produced. In an RGB LED, various colour combinations generate multiple coloured lights, and through interfacing with the OLED, the name of the colour is displayed on the screen. Table 2 shows the colour combinations of an RGB LED.

Circuit and working

Fig. 2 shows the circuit of Colour Buddy. It is built around an Arduino Uno and powered by a 9V battery. It controls a common-cathode RGB LED and an OLED display. The battery connects to the Arduino's

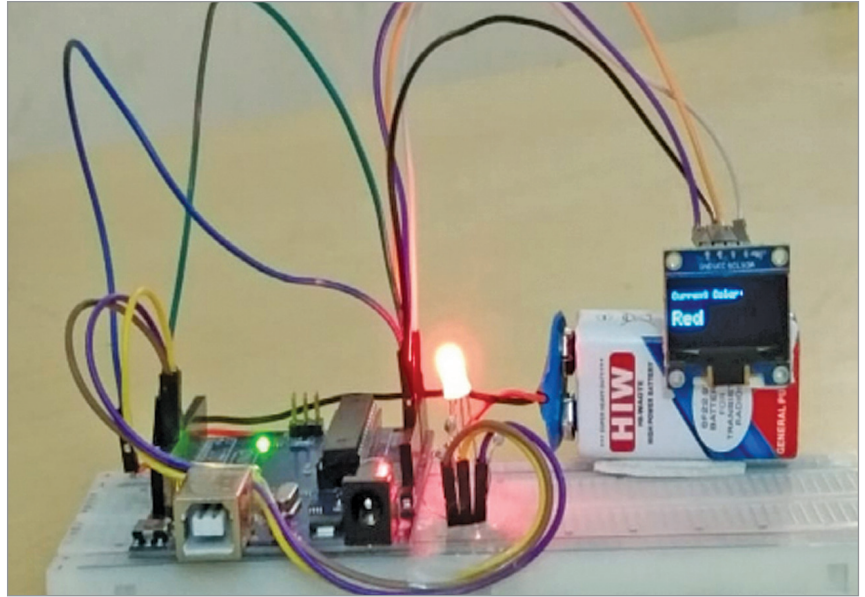


Fig. 1: Author's prototype

DC jack, which regulates the voltage to 5V.

Each LED colour pin (red, green, blue) is linked to a digital output through a 220Ω resistor to limit current. Using high/low logic signals or PWM, the Arduino mixes colours, changing the LED colour every 5 seconds while updating the OLED with the corresponding colour name.

An RGB LED contains three individual LEDs—red, green, and blue—packaged together in a single unit. Each LED is connected to the Arduino through a separate 220Ω current-limiting resistor: the red LED to pin 9, the green LED to pin 10, and the blue LED to pin 11. By adjusting the brightness of each colour using PWM, a diverse set of colours can be generated.

Another component used in this system is the 2.4cm (0.96-inch) OLED display (I²C, SSD1306), connected to the Arduino via the I²C

interface. The VCC pin is connected to the Arduino's 5V pin, GND to GND, SDA (Serial Data) to analogue pin A4, and SCL (Serial Clock) to analogue pin A5, following the standard I²C pin configuration on the Arduino Uno.

Its operation involves the Arduino controlling both the RGB LED and the OLED simultaneously. The OLED can display messages, sensor readings, or system status, while the RGB LED provides visual

TABLE 1
BILL OF MATERIALS

Name	Quantity
Arduino Uno board	1
Common-cathode RGB LED	1
OLED display (SSD1306)	1
220Ω resistor	3
9V battery with connector	1
USB cable	For programming
Connecting wires	As required